# Week 10 Solution

NPTEL PSOSM Course (Privacy and Security in Online Social Media)

**Paper for Questions 1 to 3:** *"Beware of What You Share: Inferring Home Location in Social Networks"*

Idea of the paper: even when users hide their home address, the location information they and their friends share publicly (home city, places lived, venues, geo-tagged posts) can be combined to infer where they live. The authors measure how much and how precise such location data is on several networks.

## Question 1

Which statement(s) is/are true? (multiple correct)

**(a) The authors perform a large-scale inference study in three popular social networks: Foursquare, Google+ and Twitter. ✓**

**(b) The paper is looking at different social networks, not only Foursquare. ✓**

**(c) The dataset used in the paper comprises many attributes including venues, where the location must be defined filling the open text fields (limited to 30 and 127 characters, respectively), and setting a pin in the map. ✓**

(d) None of the above

### Solution

- **(a) True:** the study covers three networks, Foursquare, Google+ and Twitter, each with its own type of location data (venues and home city; places lived, address, education and employment; profile location and geo-tagged tweets).
- **(b) True:** this follows directly from (a). Unlike the earlier Foursquare-only work, this paper compares several networks.
- **(c) True:** when a Foursquare venue is created, its location is entered as free text in fields with character limits (30 and 127) plus a pin dropped on the map. Since the text is not validated, it may be noisy, which is why the authors had to check which entries were valid.

Since (a), (b) and (c) are all true, (d) is false.

**Answer:** Options (a), (b) and (c).

## Question 2

What is a valid AGI?

(a) Valid Artificial General Intelligence

**(b) Valid Ambiguous Geographical Information ✓**

(c) Valid Artificial Geographical Interface

(d) None of the above

### Solution

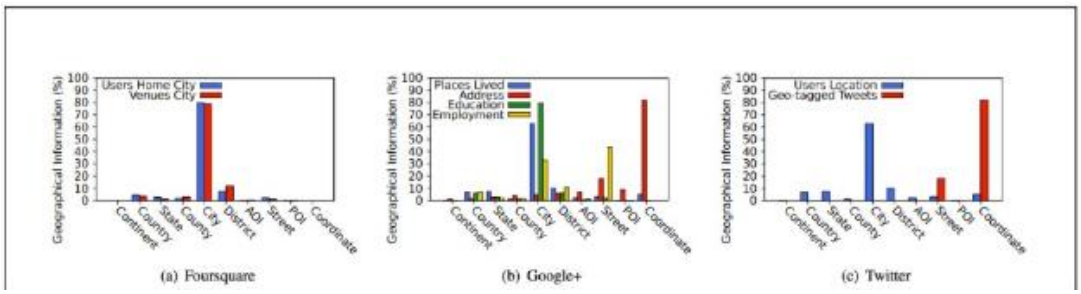
In this paper, AGI stands for Ambiguous Geographical Information: a location string that is a real place but could refer to more than one location. For example, 'Springfield' or 'Cambridge' exists in several countries, so the text alone does not say which one is meant.

A valid AGI is such an entry that does refer to a real place (as opposed to junk like 'my home' or 'Mars'), even though it needs extra context to resolve. Option (a) is a distractor that uses the more common AI meaning of AGI, which is not what the paper uses.

**Answer:** Valid Ambiguous Geographical Information.

## Question 3

Given the following figure depicts the quality of valid UGI, what is the correct inference(s) from the figure? (multiple correct)




**(a) In figure (a), the vast majority (80%) of Foursquare users and venues have location information at the city level. ✓**

(b) In figure (b), the majority of users provide home location information at country level.

(c) In figure (c), 60% of the users have provided their location information at the district level.

**(d) In figure (b), 80% of the users have provided location information at city level. ✓**

### Solution

Each panel shows, for each type of location field, the percentage of entries at each level of detail, from Continent (least precise) to Coordinate (most precise). The tallest bar shows the most common level of detail.

- **(a) True:** in Foursquare, both the blue (users' home city) and red (venues' city) bars peak at City, at about 80%.
- **(b) False:** in Google+, the Country bars are tiny (under 10%). The tall bars are at City (about 62% for Places Lived and about 80% for Education) and at Coordinate (about 80% for Address).
- **(c) False:** in Twitter, the blue (users' location) bar at about 60% is at the City level, not District. The District bar is only about 10%.
- **(d) True:** in Google+, the green (Education) bar at the City level is about 80%.

**Answer:** Options (a) and (d).

**Paper for Questions 4 to 6:** *"On the Dynamics of Username Change Behavior on Twitter"*

Idea of the paper: Twitter lets users change their username (handle) while keeping the same account. The authors tracked a large set of users over time to study how often, how and why they change handles, and the security problems this causes (broken links and username squatting).

## Question 4

From the overall conclusion of the paper, which of the statements is true?

(a) The set of people who change their handle many times is slightly larger than those who change their handle a very small number of times.

**(b) The set of people who change their handle many times is slightly smaller than the set of people who change their handle a very small number of times. ✓**

(c) The set of people who change their handle many times is roughly equal to the set of people who do not change their handle.

(d) None of the above

### Solution

Plotting the number of users against how many times they changed their username (the inset of the Question 5 figure) gives a falling curve: many users change their handle only once or a few times, and the

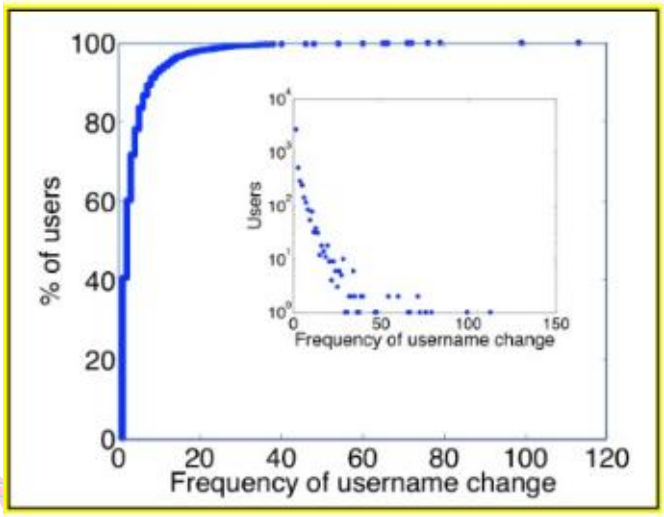
number of users drops as the number of changes grows. The paper concludes that frequent changers form the smaller group, and that this group is still sizeable rather than negligible, which is why it is described as 'slightly smaller'.

Option (a) reverses this. Option (c) is not supported, since users who never change their handle form by far the largest group on Twitter.

**Answer:** The set of people who change their handle many times is slightly smaller than the set of people who change their handle a very small number of times.

## Question 5

From the following figure, what is the correct inference?




(a) 20% of users rarely change user names in short intervals and 80% change user names frequently after longer intervals.

(b) 80% of users rarely change user names in short intervals, and 20% change user names frequently after longer intervals.

(c) 35% of users choose a user name unrelated to an old one, and 65% of users reuse an old user name.

**(d) 20% of users frequently change user names in short intervals, and 80% change user name rarely after longer intervals. ✓**

### Solution

The main plot is a CDF: the y-value is the percentage of users who changed their username at most x times. The inset shows the number of users (log scale) against how often they changed their username.

- The CDF rises almost vertically near x = 0 and reaches about 80% at only a handful of changes. So about 80% of users change rarely, with long gaps between changes.
- The remaining about 20% make up the long tail stretching to 40, 80 and even more than 100 changes. These users change frequently, which means short intervals between changes.
- The inset confirms this: the number of users drops steeply as the change frequency increases, with a few extreme users far to the right.

Options (a) and (b) mix up the pairings ('rarely' with 'short intervals', 'frequently' with 'longer intervals'), and option (c) is about how new usernames relate to old ones, which this figure does not show.

**Answer:** 20% of users frequently change user names in short intervals, and 80% change user name rarely after longer intervals.

## Question 6

What is the correct observation in the paper?

(a) This paper aims at finding how and why users change their usernames within a social network like Twitter.

(b) Most users created new usernames unrelated to the old username when they changed usernames within a social network.

(c) The authors believe that unrelated usernames over time could be credited to the absence of cognitive load to remember a past dumped username.

**(d) All of the above ✓**

**Solution**

- **(a) True:** the stated goal of the paper is to understand the frequency, patterns and reasons ('how and why') of username changes on Twitter.
- **(b) True:** comparing old and new handles with string similarity, the authors found most new usernames had little in common with the previous ones.
- **(c) True:** their explanation is that users feel no need to stay close to an old handle once they drop it. There is no cognitive load (memory effort) to keep it, so the new choice is free and often unrelated.

**Answer:** All of the above.

**Paper for Questions 7 to 10:** *"$1.00 per RT #BostonMarathon #PrayForBoston: Analyzing Fake Content on Twitter"*

Idea of the paper: after the Boston Marathon bombings, Twitter was flooded with rumours, fake charity accounts and fake images. The authors collected millions of related tweets and studied how fake content spread, which accounts spread it, and whether its spread could be predicted.

## Question 7

Which of the following statements is true about the Boston blasts?

**(a) Twin blasts occurred during the Boston Marathon on April 15th, 2013 at 18:50 GMT ✓**

(b) Twin blasts occurred during the Boston Marathon on April 15th, 2009 at 18:50 GMT

(c) Four people were killed and 264 were injured in the incident

(d) All of the above

**Solution**

- **(a) True:** two bombs exploded near the finish line on 15 April 2013, at about 2:50 pm Boston time, which is 18:50 GMT. This is the date and time given in the paper.
- **(b) False:** wrong year (2009).
- **(c) False:** the blasts killed three people and injured around 264. A fourth death (a police officer) happened days later during the manhunt, not in the blasts.
- **(d) False:** since (b) and (c) are false.

**Answer:** Twin blasts occurred during the Boston Marathon on April 15th, 2013 at 18:50 GMT.

## Question 8

What can be depicted from the tweets shared below?




(a)




(b)

(a) Both tweets contain fake content.

(b) Figure (a) depicts a tweet from a fake charity profile.

(c) Figure (b) depicts a rumour about a child being killed in the blasts.

**(d) All of the above ✓**

### Solution

- **Tweet (a):** an account named '_BostonMarathon' (not the official marathon account) promised to donate $1.00 to victims for every retweet. It was a fake charity profile: no money was donated, and the trick simply harvested over 52,000 retweets and followers. This tweet gave the paper its title.
- **Tweet (b):** @HopeForBoston posted that an 8-year-old boy who was running for the Sandy Hook kids had died. This was a false rumour: it used the emotional link to the Sandy Hook school shooting to spread rapidly.

So (b) and (c) are both correct, and together they make (a) correct: both tweets contain fake content.

**Answer:** All of the above..

## Question 9

What are the major contributions of the paper? (multiple correct)

**(a) The authors characterised the spread of fake content on Twitter using temporal, source and user attributes. ✓**

**(b) The authors used linear regression to predict how viral a rumour would be in the future based on its current user characteristics. ✓**

**(c) The authors analysed the activity and interaction graphs for the suspended user profiles created during the Boston blasts. ✓**

(d) None of the above

### Solution

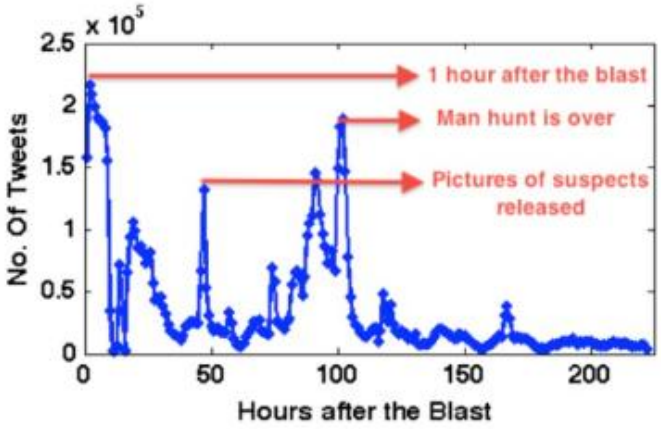
The paper lists three main contributions, matching options (a) to (c):

- **Characterisation:** how fake content spread over time (temporal), which tools and clients posted it (source), and what kind of accounts spread it (user attributes such as followers and account age).
- **Prediction:** a linear regression model that uses the impact of the users who have spread a rumour so far to predict how widely it will spread later.
- **Malicious profiles:** an analysis of accounts created after the blasts and later suspended by Twitter, studying their activity and the interaction graph among them.

**Answer:** Options (a), (b) and (c).

## Question 10

What does the following figure depict?




**(a) The figure shows the temporal distribution of tweets after the Boston blast** ✓

(b) The figure shows the spatial distribution of tweets after the Boston blast

(c) Both of the above

(d) None of the above

### Solution

The x-axis is 'Hours after the Blast' and the y-axis is the number of tweets. So the figure shows how tweet volume changed over time: a temporal distribution.

The peaks match real-world events: about 2.2 lakh (220,000) tweets in the hour right after the blast, a spike near 48 hours when the FBI released the suspects' pictures, and another spike near 100 hours when the manhunt ended.

A spatial distribution would show where tweets came from, for example on a map. The figure has no location information, so (b) and (c) are wrong.

**Answer:** The figure shows the temporal distribution of tweets after the Boston blast.

## Answer Key

| Q.No. | Correct answer |
|---|---|
| 1 | (a), (b) and (c) |
| 2 | Valid Ambiguous Geographical Information |
| 3 | (a) and (d) |
| 4 | Frequent changers are slightly fewer than rare changers |
| 5 | 20% change frequently in short intervals; 80% rarely, after longer intervals |
| 6 | All of the above |
| 7 | Twin blasts on April 15th, 2013 at 18:50 GMT |
| 8 | All of the above |
| 9 | (a), (b) and (c) |
| 10 | Temporal distribution of tweets |